PROGRAMMING ASSIGNMENT:

Think of how Turing test and Captcha could be implemented. Suggest an appropriate architecture for their design.

1. Turing Test

The **Alan Turing** proposed the **Turing Test** to determine whether a machine can exhibit intelligent behaviour similar to a human.

In this test, a **human judge communicates with a human and a machine through a text interface**. If the judge cannot reliably identify which one is the machine, the machine is said to pass the Turing Test.

The architecture consists of the following components:

1. Human Judge (Evaluator)

- Interacts with both the human and machine.
- Sends questions and receives responses through a communication interface.

2. Human Participant

- A real human answering the judge's questions.

3. AI Agent / Machine

- An intelligent program designed to respond like a human.
- Uses Natural Language Processing (NLP) and knowledge base to generate answers.

4. Communication Interface

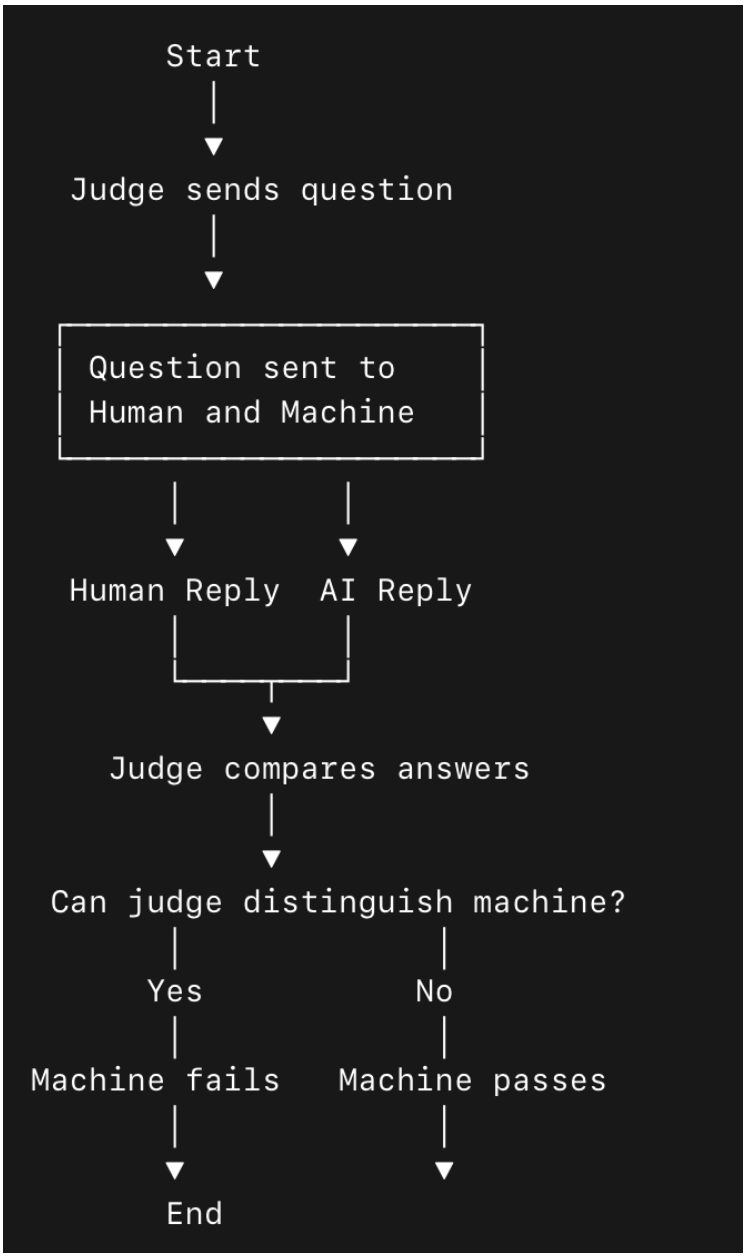
- Text-based interface that hides the identity of the participants.
- Ensures anonymous interaction.

5. Evaluation Module

- The judge analyzes the responses and decides whether they are human-like or machine-generated.

Working:

1. Judge sends questions through the interface.
2. Both human and AI agent respond.
3. Judge compares responses.
4. If the judge cannot distinguish the machine from the human, the AI passes the Turing Test.



2. CAPTCHA

CAPTCHA is a security technique used to distinguish humans from automated bots. It acts as a **reverse Turing Test**, where the computer test

The CAPTCHA system architecture contains the following components:

1. User Interface Module

- Displays the CAPTCHA challenge (distorted text, images, puzzles) to the user.

2. CAPTCHA Generator

- Automatically creates challenges that are easy for humans but difficult for bots.
- Examples: distorted characters, image recognition tasks.

3. Challenge Database

- Stores images, text patterns, or puzzles used to generate CAPTCHA challenges.

4. Response Collection Module

- Receives the user's input answer.

5. Verification Module

- Compares the user's response with the correct answer stored in the system.

6. Access Control Module

- If the response is correct → user is allowed access.
- If incorrect → a new CAPTCHA challenge is generated.

1. User requests access to a website or service.
2. System generates a CAPTCHA challenge.
3. User attempts to solve the challenge.

4. System verifies the response.
5. Access is granted only if the response is correct.

```
       Start
         |
         ▼
  User requests access
         |
         ▼
 System generates CAPTCHA
         |
         ▼
 CAPTCHA displayed to user
         |
         ▼
  User enters response
         |
         ▼
 System verifies answer
         |
     ┌───┴───┐
     ▼       ▼
  Correct   Incorrect
     |        |
     ▼        ▼
Access       Generate
Granted      New CAPTCHA
     |
     ▼
     End
```